

Research Proposal

Spectroscopic Dissection of Type Ia Supernovae for Precision Cosmological Distance Measurements

Toward Resolving the H_0 Tension and Constraining Dark Energy Evolution

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Executive Summary: Type Ia supernovae (SNe Ia) are the cornerstone of cosmological distance measurements, yet the empirical Phillips relation used to standardize their luminosities may conceal physically distinct subclasses with systematically different intrinsic brightnesses. Recent observations of “optical twins”—SN Ia pairs with nearly identical optical spectra but discrepant distance moduli—support this hypothesis, with their divergence traced to differences in near-ultraviolet (NUV) luminosity. This proposal aims to (1) develop a GPU-accelerated spectral analysis framework using **Lumina-SN** and **Lumina-ML** to extract the underlying physical parameters of SNe Ia from their spectra, (2) identify and characterize NUV-divergent subclasses through the upcoming **S-Drift** space UV survey, and (3) construct a refined absolute luminosity calibration that accounts for these subclasses, ultimately improving the precision of the Hubble constant H_0 and dark energy constraints.

1. Scientific Background and Motivation

1.1 Type Ia Supernovae as Standardizable Candles

Type Ia supernovae arise from the thermonuclear disruption of carbon–oxygen white dwarfs in binary systems. Their remarkable utility in cosmology stems from the Phillips relation (Phillips 1993): the peak luminosity of an SN Ia correlates with the rate at which its light curve declines, parameterized by $\Delta m_{15}(B)$, the magnitude drop in the B -band during the first 15 days after maximum. This single-parameter correction transforms SNe Ia from merely bright events into standardizable candles, enabling precise distance measurements across cosmological scales.

The application of this relation to distant SNe Ia led to the landmark discovery of the accelerating expansion of the universe (Riess et al. 1998; Perlmutter et al. 1999), establishing dark energy as one of the central problems in modern physics. Current SN Ia distance measurements achieve a statistical precision of $\sim 5\text{--}7\%$ per object after light-curve standardization. However, as cosmological surveys (e.g., Rubin/LSST, Roman) aim to constrain the dark energy equation of state to percent-level precision, the systematic floor of the Phillips relation itself becomes the limiting factor.

1.2 The Subclass Hypothesis: Cracks in the Phillips Relation

Key Question: Is the Phillips relation a single, universal relation, or does it conceal physically distinct subclasses of SNe Ia with systematically different luminosity calibrations?

Despite three decades of success, mounting evidence suggests that the Phillips relation may not describe a single physical sequence. Several lines of evidence point to the existence of subclasses:

- **Progenitor channel diversity:** Theoretical models predict at least two major progenitor channels—single degenerate (accretion from a companion) and double degenerate (merger of two white dwarfs)—which may produce explosions with subtly different nucleosynthetic yields and spectral properties, even when light-curve shapes are similar.
- **Residual scatter in the Hubble diagram:** After applying the Phillips relation and color corrections, the residual scatter in SN Ia distances ($\sigma \approx 0.10\text{--}0.15$ mag) exceeds what is expected from photometric errors alone, suggesting intrinsic diversity that is not captured by current standardization parameters.
- **Spectroscopic diversity:** The Wang et al. (2009) classification of SNe Ia into “Normal,” “High-Velocity” (HV), and “91T-like” / “91bg-like” groups based on Si II $\lambda 6355$ absorption velocity demonstrates that spectroscopically distinct populations exist within the nominally “normal” SN Ia class. HV SNe Ia, for example, are preferentially found in massive host galaxies and may have systematically different Hubble residuals.

1.3 Optical Twins and NUV Divergence: A Smoking Gun

Observational Evidence: “Optical twins”—SN Ia pairs with nearly identical optical spectra (3500–7500 Å) but measurably different distance moduli—have been identified. Crucially, these twins diverge in their **near-ultraviolet (NUV)** luminosity (2000–3500 Å), suggesting that the NUV carries independent information about the explosion physics that is not encoded in the optical.

The most striking observational example is the pair **SN 2011fe** and **SN 2011by** (Foley & Kirshner 2013, *ApJ*, 769, L1; Graham et al. 2015, *MNRAS*, 446, 2073). These two nearby SNe Ia exhibited nearly identical optical spectra, light-curve widths (Δm_{15}), and photometric colors throughout their photospheric and nebular phases, yet SN 2011by was ~ 0.335 mag fainter at peak in all optical bands. Crucially, HST ultraviolet spectroscopy revealed that the twins diverge dramatically below ~ 3500 Å, with SN 2011by showing significantly suppressed NUV flux (Figure 1). Foley & Kirshner argued that this NUV difference traces a factor of ~ 1.7 difference in the ^{56}Ni mass, though Graham et al. found the nebular spectra to be nearly identical, suggesting the situation is more nuanced and may involve differences in progenitor metallicity or iron-group element distribution rather than total ^{56}Ni mass alone.

Further support comes from Milne et al. (2013, *ApJ*, 779, 23; 2015, *ApJ*, 803, 20), who used Swift UVOT photometry to demonstrate that normal SNe Ia separate into two distinct NUV–optical color subgroups: a dominant “NUV-red” population and a “NUV-blue” population comprising $\sim 1/3$ of the sample, offset by ~ 0.4 mag in NUV–optical color. The NUV-blue group shows less iron-peak absorption in the 2900–3500 Å region and evidence of unburned carbon, pointing to distinct explosion physics. More recently, Burns et al. (2020, *ApJ*, 895, 118) studied the “sibling” SNe Ia 2013aa and 2017cbv in the same host galaxy NGC 5643, providing an ideal controlled comparison that eliminates systematics from distance and peculiar velocity, while Riess et al. (2024, *ApJ*, 977, 180) demonstrated that spectroscopic twins can be used toward a precise determination of H_0 .

The resolution of the twin luminosity discrepancy appears to lie in the NUV. The near-ultraviolet region ($\lambda \lesssim 3500$ Å) is dominated by a dense forest of iron-group element (IGE) absorption lines—Fe II, Fe III, Co II, Ni II—that are highly sensitive to:

1. **The total mass of ^{56}Ni synthesized in the explosion**, which directly determines the peak luminosity through radioactive decay heating ($^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$).
2. **The radial distribution of iron-group elements** within the ejecta, which controls the opacity structure and hence the emergent UV flux.
3. **The progenitor metallicity**, which sets the initial abundance of stable iron-peak elements (e.g., ^{54}Fe , ^{58}Ni).

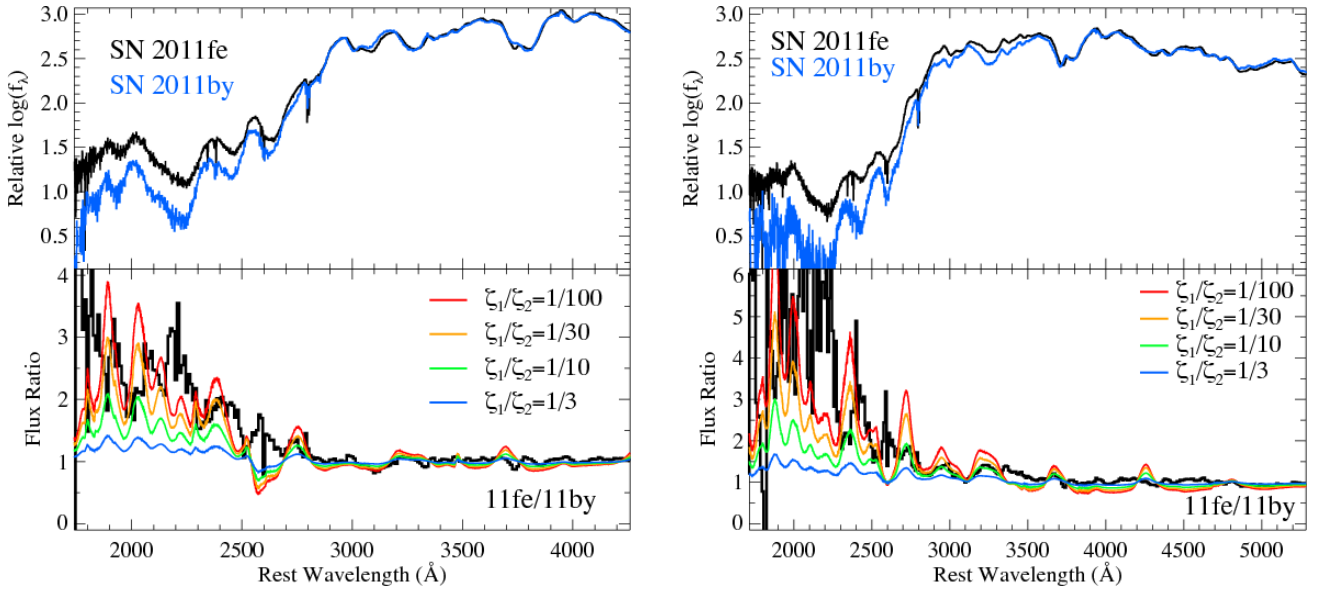


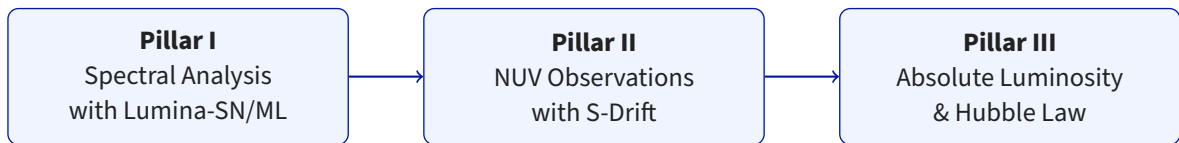
Figure 1: UV-optical spectral comparison of the optical twins SN 2011fe (black) and SN 2011by (blue) at two epochs: pre-maximum (−9/−10 days; **left**) and near-maximum (−2/−1 days; **right**). **Top panels:** The two SNe have nearly identical spectra at optical wavelengths ($\lambda > 3500 \text{ \AA}$) but diverge dramatically in the NUV ($\lambda < 3000 \text{ \AA}$). The NUV discrepancy is larger and extends to redder wavelengths at earlier epochs. **Bottom panels:** Flux ratio (SN 2011fe / SN 2011by); colored lines show predictions for different progenitor metallicity ratios (ζ_1/ζ_2) from Lentz et al. (2000). (Figures from Graham et al. 2015, *MNRAS*, 446, 2073.)

that contribute to UV line blanketing without affecting the radioactive energy budget.

Thus, two SNe Ia can have identical optical spectra (governed primarily by intermediate-mass elements like Si and S in the outer ejecta) while differing in their NUV flux (governed by iron-group elements concentrated in the inner ejecta). The NUV flux ratio between optical twins encodes precisely the information needed to break the degeneracy and determine the absolute luminosity.

2. Proposed Research Program

This proposal describes a three-pronged research program that connects theoretical spectral modeling, observational NUV surveys, and cosmological distance measurement:



2.1 Pillar I: GPU-Accelerated Spectral Analysis with Lumina-SN and Lumina-ML

The first pillar leverages two codes that I have developed: **Lumina-SN**, a C/CUDA Monte Carlo radiative transfer code, and **Lumina-ML**, a neural spectral emulation and Bayesian inference framework.

Lumina-SN: The Forward Model

Lumina-SN solves the 1D Monte Carlo radiative transfer problem in homologously expanding SN Ia ejecta using the Sobolev approximation. The code models:

- Photon packet emission from a sharp inner photosphere at velocity v_{ph} , propagation through radial shells with a broken power-law density profile, and interactions with atomic species via Thomson scattering and bound-bound (Sobolev) line opacity.
- Iterative convergence of the radiation field (dilution factor W , radiation temperature T_{rad}) with the plasma

state (LTE ionization/excitation) over 5–20 cycles.

- A three-zone ejecta structure (Core, Wall, Outer) with independently tunable density exponents and composition (Si, Fe, O, C, Ca, S, Mg, Ni mass fractions).

The C/CUDA implementation achieves $\sim 10\times$ speedup over the Python-based TARDIS code on NVIDIA GPUs, enabling the rapid evaluation of tens of thousands of forward models required for parameter inference.

Lumina-ML: Neural Emulation and Bayesian Inference

Even with GPU acceleration, a single Lumina-SN forward evaluation requires ~ 1 second. To enable full Bayesian inference over a 15-dimensional parameter space, Lumina-ML constructs a **neural spectral emulator**:

1. **Training set generation:** 50,000 Lumina-SN spectra are computed on a Latin Hypercube Sample spanning the 15-dimensional parameter space (luminosity, v_{ph} , density parameters, elemental mass fractions for the three ejecta zones).
2. **PCA compression:** Each synthetic spectrum ($\sim 1,100$ wavelength bins) is projected onto ~ 60 principal components, capturing $>99.9\%$ of the spectral variance.
3. **Neural network:** A multilayer perceptron (15 inputs $\rightarrow$ 60 PCA coefficients, $\sim 500\text{K}$ parameters, SiLU activations) is trained to map physical parameters to PCA-compressed spectra in $\sim 10\ \mu\text{s}$ per evaluation—five orders of magnitude faster than the full simulation.
4. **Bayesian inference:** Three complementary methods are implemented:
 - MCMC with `emcee` for posterior sampling
 - Nested sampling with `dynesty` for evidence computation and model comparison
 - Simulation-based inference (SBI/SNPE) for amortized neural posterior estimation

Application to the twin problem: Given an observed SN Ia spectrum, Lumina-ML returns the full posterior distribution over 15 physical parameters. For an optical twin pair, we can quantify *exactly* which parameters differ—e.g., whether the NUV discrepancy is driven by the total ^{56}Ni mass, the iron-group distribution, or the progenitor metallicity. This physical diagnosis is the key to constructing an improved luminosity calibration.

2.2 Pillar II: NUV Observations with S-Drift

The near-ultraviolet (2000–3500 Å) is inaccessible from the ground due to atmospheric absorption, requiring space-based observations. The **S-Drift** project—a planned UV space telescope—provides an ideal platform for systematic NUV photometry and spectroscopy of nearby SNe Ia.

The observational program will:

- Obtain NUV photometry ($\sim 2500\ \text{\AA}$) and low-resolution spectroscopy for a volume-limited sample of SNe Ia ($z < 0.05$) discovered by ground-based surveys (ZTF, Rubin/LSST).
- Identify optical twin pairs from ground-based spectral databases and measure their NUV flux ratios.
- Construct the first statistically significant NUV-optical color–luminosity relation, parameterizing the NUV excess/deficit relative to the optical twin match.
- Cross-correlate NUV properties with host galaxy characteristics (mass, metallicity, star formation rate) to test the progenitor metallicity hypothesis.

2.3 Pillar III: Refined Absolute Luminosity and the Hubble Law

The ultimate goal is to construct a **subclass-aware luminosity calibration** that replaces the single Phillips relation with a multi-parameter standardization incorporating NUV information:

$$M_B = M_0 + \alpha \cdot (\Delta m_{15} - 1.1) + \beta \cdot (B - V)_{\max} + \gamma \cdot \Delta_{\text{NUV}} + \delta \cdot \theta_{\text{Fe}} \quad (1)$$

where Δ_{NUV} is the NUV color excess (measured by S-Drift) and θ_{Fe} is the iron-group parameter derived from Lumina-ML spectral fitting. The inclusion of these two additional parameters is expected to reduce the Hubble diagram residual scatter from $\sigma \approx 0.12$ mag to $\sigma \lesssim 0.08$ mag, corresponding to a $\sim 30\%$ improvement in individual distance precision.

This improved calibration will be applied to:

- **The Hubble constant H_0 :** A more precise SN Ia distance ladder tightens the local H_0 measurement, directly addressing the Hubble tension ($H_0 = 73.0 \pm 1.0$ km/s/Mpc from SNe vs. 67.4 ± 0.5 km/s/Mpc from CMB).
- **Dark energy equation of state $w(z)$:** Reduced systematic scatter in the Hubble diagram improves constraints on the time evolution of dark energy, testing whether $w = -1$ (cosmological constant) or $w(z)$ varies with redshift.

3. Methodology and Timeline

Phase 1: Spectral Template Library Construction (Year 1)

- Generate a comprehensive grid of $\sim 100,000$ Lumina-SN synthetic spectra spanning the full SN Ia parameter space, with emphasis on the iron-group abundance and distribution parameters that control NUV flux.
- Train and validate the Lumina-ML neural emulator on this expanded grid, achieving spectral reconstruction accuracy of $< 1\%$ RMS across the UV-optical wavelength range (2000–10000 Å).
- Apply the Bayesian inference pipeline to archival SN Ia spectra from the literature (CfA, CSP, Berkeley samples) to establish baseline physical parameter distributions.

Phase 2: Optical Twin Identification and NUV Prediction (Year 1–2)

- Develop an automated optical twin identification algorithm based on spectral cross-correlation in the 3800–7200 Å range, applied to the existing spectral databases ($\sim 3,000$ SN Ia spectra near maximum light).
- For each identified twin pair, use Lumina-ML to infer the posterior distributions of all 15 physical parameters and identify the parameters responsible for the luminosity discrepancy.
- Generate *predictions* for the NUV flux difference within each twin pair, to be verified by S-Drift observations.

Phase 3: S-Drift NUV Observations and Calibration (Year 2–3)

- Coordinate with the S-Drift mission team to obtain NUV photometry and spectroscopy of newly discovered SNe Ia and known twin candidates.
- Validate the Lumina-ML predictions against observed NUV fluxes.
- Construct the subclass-aware luminosity calibration (Equation 1) and quantify the improvement in Hubble diagram scatter.

Phase 4: Cosmological Application (Year 3)

- Apply the refined calibration to the Pantheon+ and Union SN Ia compilations to derive updated constraints on H_0 and $w(z)$.
- Assess whether the subclass correction alleviates the Hubble tension.
- Prepare the calibration framework for application to the Rubin/LSST and Roman SN Ia samples.

4. Technical Readiness and Resources

- **Lumina-SN** is fully operational, validated against TARDIS to sub-1% accuracy (W error 0.44%, T_{rad} error 0.36%), and demonstrated on SN 2011fe (best-fit RMS = 0.0894). The C/CUDA implementation achieves $\sim 10\times$ GPU speedup (200K packets: 728 ms on GPU vs. 7.2 s on CPU).
- **Lumina-ML** provides a complete neural emulation and Bayesian inference pipeline with three inference backends (MCMC, nested sampling, SBI). The emulator evaluates spectra in $\sim 10\ \mu\text{s}$, enabling exhaustive parameter space exploration.
- **Computing resources:** KIAS Center for Advanced Computation provides dedicated GPU clusters (NVIDIA A100/H100). Additional allocations on the KISTI Nurion/Neuron supercomputers are available through the Grand Challenge Program.
- **Collaboration network:** Established collaborations with the KASI supernova group (Korea Astronomy and Space Science Institute), the FOSTER group for multi-wavelength observations, and the SDSS Korean Scientist Group for spectral databases.

5. Expected Outcomes and Impact

1. **Physical classification of SN Ia subclasses:** The first systematic mapping between spectroscopically inferred physical parameters (iron-group mass, distribution, progenitor metallicity) and the optical twin / NUV divergence phenomenon.
2. **NUV-augmented luminosity calibration:** A new standardization relation (Equation 1) that reduces Hubble diagram scatter by $\sim 30\%$, from $\sigma \approx 0.12$ to $\sigma \lesssim 0.08$ mag per SN.
3. **Improved H_0 measurement:** Tighter local distance ladder constraints directly bearing on the Hubble tension, one of the most pressing problems in modern cosmology.
4. **Public code release:** Lumina-SN and Lumina-ML, along with the trained neural emulator and spectral template library, will be publicly released for use by the SN Ia community.
5. **S-Drift synergy:** This program provides a key science case for the S-Drift UV space mission, demonstrating the cosmological value of systematic NUV observations of SNe Ia.

6. Cosmological Applications: the H_0 Tension and Dynamic Dark Energy

The refined SN Ia luminosity calibration developed in this program has direct implications for two of the most pressing open questions in cosmology:

6.1 The Hubble Tension

The $\sim 5\sigma$ discrepancy between the local measurement of the Hubble constant ($H_0 = 73.0 \pm 1.0$ km/s/Mpc from the SH0ES SN Ia distance ladder; Riess et al. 2022) and the value inferred from the cosmic microwave background ($H_0 = 67.4 \pm 0.5$ km/s/Mpc from Planck; Planck Collaboration 2020) remains unresolved. A significant fraction of the local H_0 error budget originates from the SN Ia standardization step. If unrecognized SN Ia subclasses with systematically different luminosities are mixed in the calibrator sample, the resulting H_0 measurement could be biased. Our NUV-augmented calibration will:

- Quantify whether the current SH0ES calibrator sample contains a mixture of NUV-red and NUV-blue subclasses, and if so, whether correcting for this mixture shifts H_0 .
- Provide a cleaner distance ladder using subclass-homogeneous SN Ia samples, reducing systematic uncertainties in H_0 to below the current 1.4% level.

- Test whether the Hubble tension is partially an artifact of SN Ia astrophysical systematics or requires new physics beyond Λ CDM.

6.2 Dynamic Dark Energy and the Equation of State $w(z)$

Recent results from the DESI BAO survey (DESI Collaboration 2024) have provided tantalizing hints that the dark energy equation of state may evolve with redshift, with $w(z)$ potentially crossing -1 at $z \sim 0.5$. Independently, our KIAS group’s analysis of the tomographic Alcock–Paczyński test (Dong et al. 2023, *ApJ*, 953, 98) found evidence for $w > -1$ from SDSS data. Confirming or refuting dynamic dark energy requires percent-level precision in the SN Ia Hubble diagram at $0.01 < z < 1.0$.

Our program contributes to this goal in two ways:

- **Reduced scatter at low redshift ($z < 0.1$):** The NUV-corrected calibration reduces the per-SN distance error from $\sim 7\%$ to $\sim 4\%$, directly tightening constraints on the anchor of the distance–redshift relation.
- **Redshift-dependent subclass fractions:** Milne et al. (2015) showed that the ratio of NUV-blue to NUV-red SNe Ia changes with redshift, which, if uncorrected, introduces a redshift-dependent systematic in the Hubble diagram that can mimic or mask dynamic dark energy. Our subclass-aware calibration removes this systematic, enabling a clean test of $w(z) \neq -1$.

By combining the improved SN Ia distances with BAO measurements from DESI/SDSS and CMB constraints from Planck/ACT, we will construct a joint cosmological analysis that simultaneously constrains H_0 , Ω_m , and the CPL dark energy parameters (w_0, w_a) . This multi-probe approach will provide the most stringent test to date of whether dark energy is truly a cosmological constant or an evolving component of the universe.

In summary, this proposal bridges computational spectral modeling (Lumina-SN/ML), space-based NUV observations (S-Drift), and precision cosmology to address a fundamental limitation in our use of Type Ia supernovae as cosmological distance indicators. By revealing and quantifying the physical substructure hidden within the Phillips relation, we aim to improve SN Ia distance precision by $\sim 30\%$, directly contributing to the resolution of the H_0 tension and to constraining the time evolution of dark energy.